RecruitIQ

Talent Acquisition Through AI

• Bias-Free Ranking • Real-Time HR Intelligence

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GitHub Repository: https://github.com/ayushi18-agarwal/RecruitIQ

1. Problem Context

The global recruitment industry processes over 1 billion job applications annually, yet studies consistently show that human recruiters spend 23% of their working time on manual resume screening — a task that is repetitive, inconsistent, and heavily influenced by unconscious bias. Traditional ATS tools filter by keyword matching alone, routinely rejecting qualified candidates whose resumes don't use the exact terminology a recruiter thought to specify. Meanwhile, strong candidates apply and wait weeks for feedback, only to learn that their application was buried under hundreds of others.

RecruitIQ is an AI-powered Recruitment Intelligence Platform that addresses these challenges by:

- **Improving candidate screening** through semantic resume analysis, skill matching, and predictive fit scoring.
- **Accelerating hiring decisions** with real-time alerts for high-potential candidates.
- **Enhancing interview readiness** through AI-generated, candidate-specific interview questions.
- **Providing a unified recruitment environment** that integrates job posting, screening, ranking, notifications, and interview preparation within a single platform.

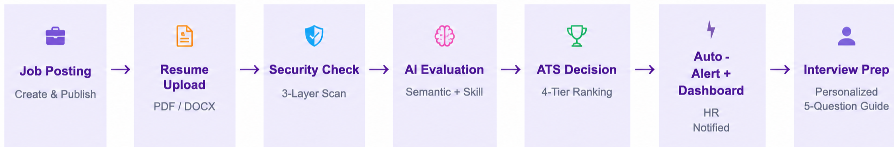
2. Objectives

- To automate resume screening and candidate ranking using AI-driven evaluation.
- To improve candidate-job matching through semantic analysis and skill-based scoring.
- To reduce manual effort and accelerate recruitment decision-making.
- To generate personalized interview questions based on candidate profiles and skill gaps.
- To provide real-time recruitment insights, alerts, and analytics for recruiters.

3. SDG Mapping — Sustainable Development Goals

SDG Goal	How RecruitIQ Aligns
SDG 4 – Quality Education	The skill-gap heatmap and personalized AI interview guides help candidates and institutions understand exactly which technical skills are in demand, enabling more targeted learning and upskilling.
SDG 8 – Decent Work & Economic Growth	By automating and democratizing access to structured, unbiased candidate evaluation, RecruitIQ accelerates fair hiring and helps organizations build stronger, more productive workforces.
SDG 9 – Industry, Innovation & Infrastructure	The platform demonstrates practical application of AI/ML in enterprise HR infrastructure — reducing operational overhead and building resilient, data-driven recruitment systems.
SDG 10 – Reduced Inequalities	Rule-based, score-driven shortlisting removes unconscious bias from early-stage screening. Every candidate is evaluated on the same objective criteria — skills, experience fit, and semantic alignment — not subjective impressions.
SDG 17 – Partnerships for the Goals	LinkedIn syndication and Zapier webhook integration enable RecruitIQ to connect with broader professional ecosystems, expanding reach to diverse candidate pools across geographies.

4. Core Workflow



Complete pipeline: Job posting → Resume upload → Security check → AI evaluation → ATS decision → Auto HR alert → Personalized interview guide

5. Methodology & Approach

- **Intelligent Job Posting & Distribution:** Creates and automatically publishes job openings across integrated platforms, including LinkedIn.
- **Secure Resume Processing:** Validates, analyzes, and securely stores uploaded resumes.
- **Hybrid AI Scoring Engine:** Evaluates candidates using semantic matching, skill analysis, predictive scoring, and experience assessment.
- **Duplicate Detection:** Identifies duplicate and potentially fraudulent applications.
- **Audit Trail & Transparency:** Maintains secure logs of recruitment activities and AI-driven decisions.
- **AI-Powered Interview Preparation:** Generates personalized interview questions based on candidate skills and skill gaps.
- **Voice-Enabled HR Assistant:** Provides voice-based access to recruitment insights and candidate information.
- **Automated Recruiter Alerts:** Sends real-time notifications for high-potential candidates.
- **Output Classification:** Categorizes candidates as Strong Hire, Hire, Review, or Reject.

6. System Architecture

Layer	Technical Detail
Document Extraction	PDF → pdfplumber (page-wise text) • DOCX → python-docx (paragraphs + tables) • Regex-based email extraction from resume body
Security Pipeline	Layer 1: Extension whitelist (.pdf, .docx only) • Layer 2: Magika ML binary content scan (detects renamed executables) • Layer 3: UUID-prefixed sandboxed storage + server-side path traversal guard
AI Scoring Engine	4-signal hybrid model: 35% MiniLM semantic similarity (all-MiniLM-L6-v2, 384-dim embeddings) + 35% alias-aware skill coverage + 30% Ridge regression (trained on 9,846 labelled resumes across 34 domains) + up to +10 pts experience bonus
Duplicate Detection	Email + job-id exact match • Cross-role email match • MD5 fingerprint of first 120 resume tokens — catches same person applying under different names
Interview Prep AI	Groq LLaMA 3.3 70B • Candidate-specific prompt (name, proven skills, exact gaps, full JD) • JSON-structured 5-question output • Personalized rule-based fallback if AI unavailable
Candidate Chatbot	LLaMA 3.3 70B • Job-context injected per message • Endpoint completely isolated from HR pipeline data
Voice Intelligence	Web Speech API (client-side, no external service) • Keyword router for nav + data commands • /api/voice/ask: open Q&A with live DB context in LLM system prompt
Alert Engine	smtplib SMTP/TLS • Fires on new Strong Hire (not on re-apply) • Dual-path: self-apply + bulk upload • Non-blocking: runs after DB commit, never delays candidate confirmation
Backend + Auth	Flask + SQLite • Bcrypt hashed passwords • CSRF token on every form • Session lockout after 5 failed attempts • Role-based route guards

7.Key Features

- **Multi-layer resume security:** with file validation and secure storage.
- **Hybrid AI scoring:** using semantic matching, skill analysis, and experience evaluation.
- **Duplicate and fraud detection:** through application validation and resume fingerprinting.

- **AI-powered candidate chatbot:** for application guidance and support.
- **Personalized interview question generation:** based on candidate profiles and skill gaps.
- **Real-time recruiter alerts:** for high-potential candidates.
- **Voice-enabled HR assistant:** for recruitment insights and navigation.
- **Transparent application tracking:** with candidate status updates.
- **Recruitment analytics and skill-demand insights:** for data-driven hiring.
- **Automated LinkedIn job posting:** through platform integration.

8.Technologies Used

Category	Technologies
Backend Framework	Python 3, Flask, SQLite
AI / ML – Scoring	all-MiniLM-L6-v2 (sentence-transformers), Ridge Regression + Logistic Regression (scikit-learn), trained on 9,846 samples across 34 job domains
AI / ML – Language	Groq API, LLaMA 3.3 70B (interview prep, chatbot, voice Q&A)
Resume Parsing	pdfplumber (PDF), python-docx (DOCX)
File Security	Magika — Google's ML-based binary file-type detector
Voice Interface	Web Speech API (SpeechRecognition + SpeechSynthesis)
Email Alerts	Python smtplib, SMTP/TLS (compatible with Gmail, SendGrid, Mailgun)
Authentication	Flask sessions, Werkzeug bcrypt hashing, CSRF tokens, login lockout
Frontend	Jinja2 templates, Chart.js, Vanilla JS, dark/light theme system
External Integrations	Zapier Webhook (LinkedIn job syndication)

9. Expected Outcome

- Reduce resume screening time through automated AI-based candidate evaluation and ranking.
- Enable faster hiring decisions with ranked shortlists, fit scores, and skill-gap insights.
- Improve interview effectiveness through AI-generated, candidate-specific interview preparation.
- Promote fair and objective hiring through skill-based and semantic candidate assessment.
- Increase visibility of qualified candidates through real-time screening and recruiter alerts.
- Expand employment opportunities through automated job distribution across professional platforms.

10.Future Scope

1. **Multi-Tenant SaaS Architecture:** Enable multiple organizations to use RecruitIQ through isolated workspaces, making the platform suitable for enterprise-scale deployment.
2. **Advanced AI Scoring — Multimodal Resume Understanding:** Integrate vision-language models to analyze resume layout, formatting, portfolios, and visual elements beyond plain text.
3. **Automated Video Interview Analysis:** Evaluate candidate video responses using speech-to-text and LLM-based assessment to automate interview evaluation.
4. **Real-Time Candidate Ranking Updates:** Provide candidates and recruiters with transparent score breakdowns, skill-gap explanations, and personalized improvement recommendations.
5. **Explainable AI (XAI) Score Breakdown for Candidates:**Provide candidates and recruiters with transparent score breakdowns, skill-gap explanations, and personalized improvement recommendations.
6. **Integration with External Job Boards and HRMS:** Integrate with platforms such as LinkedIn, Naukri, Workday, and SAP SuccessFactors for seamless recruitment workflow automation.
7. **Multilingual Resume Support:** Support resumes in multiple languages through language detection and translation, enabling wider accessibility across diverse talent pools.